

Speech Emotion Recognition Using Librosa

AIM: To study and understand how speech emotions can be recognized from audio signals using the Librosa library, and to analyze different audio features like MFCC, spectrogram, and mel-spectrogram.

THEORY:

Speech Emotion Recognition (SER) is a technique used to detect human emotions from speech signals. Emotions like happy, sad, angry, fearful, etc., can be identified using machine learning and signal processing methods.

According to the uploaded paper, SER plays an important role in:

- Human-computer interaction
- Healthcare (mental health analysis)
- Entertainment systems

LIBRARIES USED:

Librosa Library:

Librosa is a Python library used for audio and music analysis. It helps in loading audio files, extracting features, and visualizing waveforms and spectrograms. It is widely used in speech processing, sound recognition, and music analysis.

Pandas Library:

Pandas is used for handling and organizing data. It helps convert raw and unstructured data into a clean and readable format, making it easier to process and analyze.

Visual Representation Techniques

Spectrogram:

A spectrogram is a visual representation of sound that shows frequency vs time. It is created using the Fourier Transform, which breaks the audio signal into different frequencies and shows their strength. It helps in understanding patterns in audio signals.

ROLE OF LIBROSA:

Librosa is a Python library used for audio analysis. It helps in:

- Loading audio files, Extracting features, Visualizing audio signals

IMPORTANT AUDIO FEATURES:

1. SPECTROGRAM

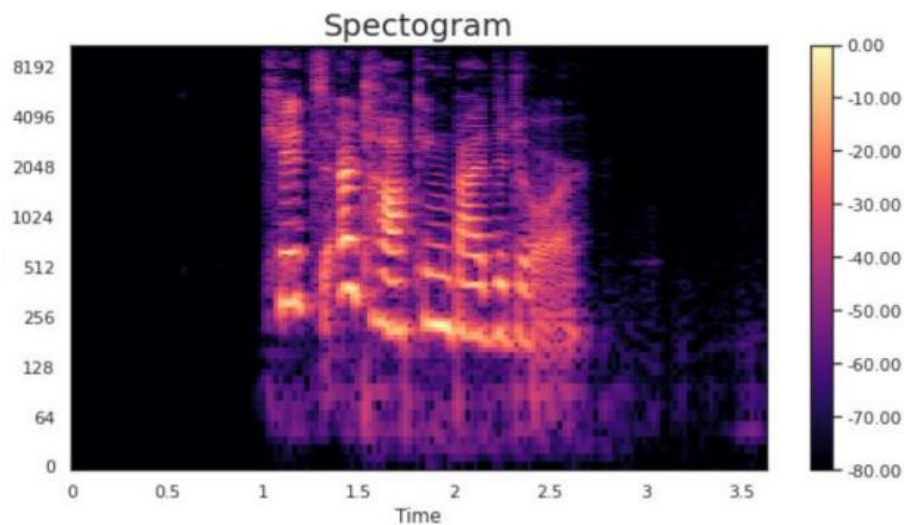


Fig.1: Plot of Spectrogram

A spectrogram is a visual representation of an audio signal that shows how frequency changes over time. It converts sound into a 2D image where:

- The x-axis represents time
- The y-axis represents frequency
- The color or intensity represents amplitude (loudness)

It is created using a mathematical method called the Fourier Transform, which breaks the audio signal into its frequency components.

Spectrograms are useful because:

- They help us understand patterns in speech signals
- Different emotions show different frequency patterns
- It makes it easier to analyze complex audio visually

In speech emotion recognition, spectrograms help in identifying variations in tone, pitch, and energy of speech.

2. MEL-SPECTROGRAM

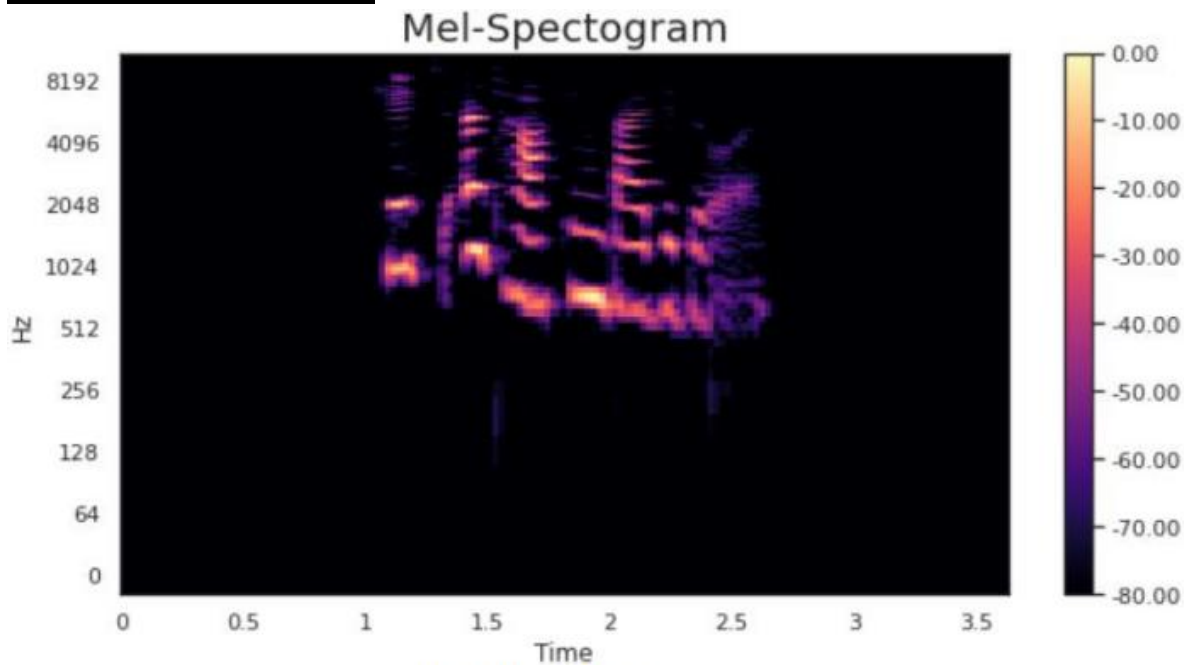


Fig.2: Plot of Mel-Spectrogram

A Mel-Spectrogram is an improved version of a normal spectrogram. It is based on the Mel scale, which is designed to match how humans actually hear sound.

Unlike a normal spectrogram:

- It gives more importance to lower frequencies
- It compresses higher frequencies

This makes it more suitable for tasks like:

- Speech recognition
- Emotion detection
- Machine learning models

Advantages of Mel-Spectrogram:

- Closer to human hearing perception
- Reduces unnecessary data
- Improves model performance

Because of these reasons, it is widely used in modern audio-based machine learning systems.

3. MFCC (Mel Frequency Cepstral Coefficients)

MFCC is one of the most important and widely used features in speech processing.

It is derived from the Mel-Spectrogram and represents the short-term power spectrum of sound.

In simple terms:

- It captures the unique characteristics of a voice signal
- It focuses on how humans perceive sound

MFCC helps in:

- Identifying speaker emotions
- Recognizing speech patterns
- Improving classification accuracy

Key points:

- Usually 13–40 coefficients are extracted
- It removes noise and irrelevant information
- It is compact and efficient

Because of its effectiveness, MFCC is considered the core feature in most speech emotion recognition systems.

4. CHROMA FEATURES

Chroma features represent the pitch content of an audio signal.

In music and speech:

- Sound can be divided into 12 pitch classes (like musical notes)
- Chroma groups frequencies into these classes

Important aspects:

- It ignores octave differences
- Focuses only on pitch similarity

Uses:

- Music analysis
- Tone detection

- Emotion recognition

In speech emotion recognition:

- Helps detect variations in pitch
- Useful for identifying emotions like happy, angry, or sad

WORKING PRINCIPLE:

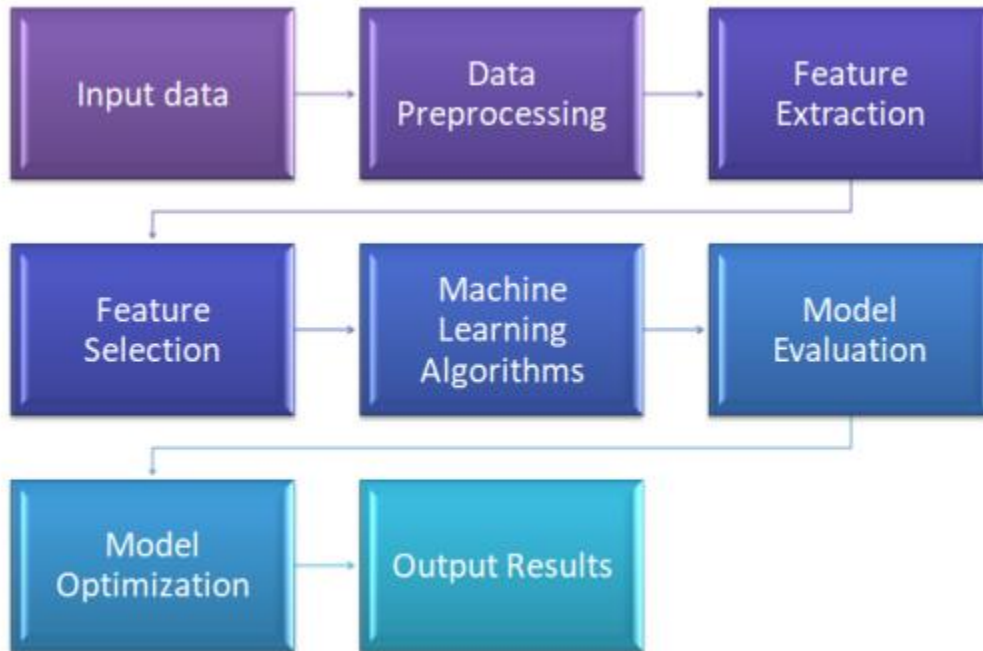


Fig.14: Working Block Diagram

The flow is:

- Input audio
- Preprocessing
- Feature extraction
- Machine learning
- Output (emotion)

CODE:

```
import librosa
import librosa.display
import numpy as np
import matplotlib.pyplot as plt

# load audio
file_path = "focusmix.wav.wav"
y, sr = librosa.load(file_path)

print("Sample Rate:", sr)
print("Audio Length:", len(y))

# waveform
plt.figure(figsize=(10, 4))
plt.plot(y)
plt.title("Waveform")
plt.xlabel("Time")
plt.ylabel("Amplitude")
plt.show()

# spectrogram
D = librosa.stft(y)
S_db = librosa.amplitude_to_db(np.abs(D), ref=np.max)

plt.figure(figsize=(10, 4))
librosa.display.specshow(S_db, sr=sr, x_axis='time', y_axis='log')
```

```

plt.title("Spectrogram")
plt.colorbar()
plt.show()

# mel spectrogram
mel = librosa.feature.melspectrogram(y=y, sr=sr)
mel_db = librosa.power_to_db(mel, ref=np.max)

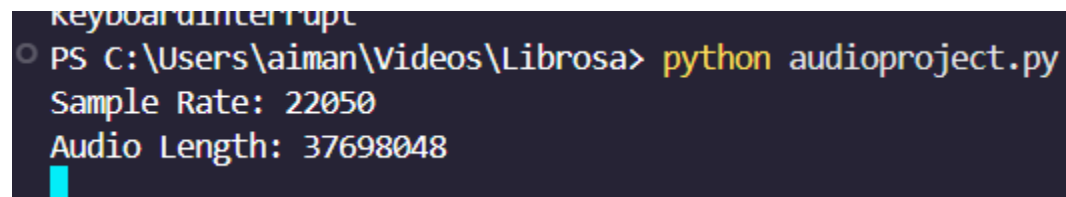
plt.figure(figsize=(10, 4))
librosa.display.specshow(mel_db, sr=sr, x_axis='time', y_axis='mel')
plt.title("Mel Spectrogram")
plt.colorbar()
plt.show()

# features
mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=40)
chroma = librosa.feature.chroma_stft(y=y, sr=sr)

print("MFCC shape:", mfcc.shape)
print("Chroma shape:", chroma.shape)
print("Mel shape:", mel.shape)

```

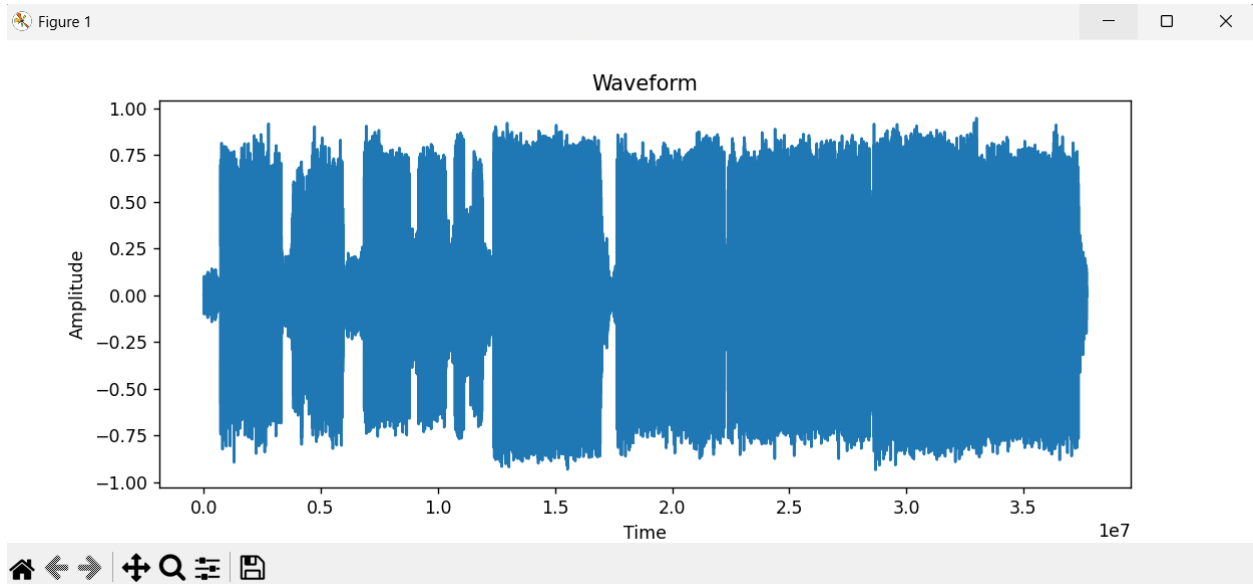
OUTPUT



```

KeyboardInterrupt
PS C:\Users\aiman\Videos\Librosa> python audioproject.py
Sample Rate: 22050
Audio Length: 37698048

```



RESULT

The system is able to extract useful features from speech signals and use them to identify emotions.

According to the paper, the model achieved around **78% accuracy** in emotion prediction.

DISCUSSION

- Audio signals are complex and vary due to:
 - Noise
 - Speaker differences
 - Environment
- Feature extraction is the most important step
- Librosa makes processing easy and efficient
- Accuracy depends on:
 - Dataset quality
 - Feature selection
 - Model used

FUTURE SCOPE:

Speech Emotion Recognition (SER) has strong potential for future development and improvement. One major direction is the use of advanced deep learning models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. These models are capable of learning complex patterns in audio data and can significantly improve the accuracy of emotion detection.

Another important area is noise reduction and signal enhancement. Real-world audio often contains background noise and distortions, which affect model performance. Developing better preprocessing techniques can help improve reliability and robustness.

The implementation of real-time emotion detection systems is also a promising direction. Such systems can analyze speech instantly and provide immediate feedback, which can be useful in applications like customer service, virtual assistants, and interactive systems.

Furthermore, SER can be widely applied in AI-based assistants and mental health monitoring tools. It can help in detecting emotional states such as stress, anxiety, or depression, enabling better human-computer interaction and support systems.

LITERATURE REVIEW:

Based on the research paper, several studies have explored the use of Librosa for feature extraction in Speech Emotion Recognition. Librosa is commonly used to extract important audio features such as MFCC, chroma, and mel-spectrogram, which are then used for classification tasks.

Various machine learning and deep learning techniques have been applied in this domain. These include:

- **Convolutional Neural Networks (CNN)** for learning spatial features from spectrograms
- **Support Vector Machines (SVM)** for effective classification of emotions
- **Multilayer Perceptron (MLP)** classifiers for basic neural network-based prediction

The **RAVDESS dataset** is one of the most widely used datasets in SER research. It contains labeled audio samples with different emotions, which helps in training and evaluating models effectively.

Research findings indicate that **accuracy improves with better feature extraction and selection techniques**. Combining multiple features and using advanced models leads to more reliable results. However, challenges such as noise, speaker variability, and data limitations still remain areas for further research.

REFERENCE

1. Z. Tang et al., “Speech Emotion Recognition using CNN and Librosa”, IEEE, 2020
2. F. Liu et al., “MFCC + SVM using Librosa”, ICIM, 2020
3. S. Saha et al., “SER using Librosa and SVM”, IEEE, 2021
4. V. P. N et al., “CNN with Librosa Features”, 2021
5. N. Kaur, H. Gupta, “Deep Learning with Librosa”, 2021
6. Librosa Documentation – <https://librosa.org>
7. Kaggle – RAVDESS Dataset
8. sklearn Documentation
9. Python Official Docs
10. NumPy Documentation
11. Matplotlib Documentation
12. Speech Processing Basics – IEEE Papers